



Math for Economists Bootcamp

Module 2, Part 2: Constrained Optimization, Homogeneity and Other Theorems

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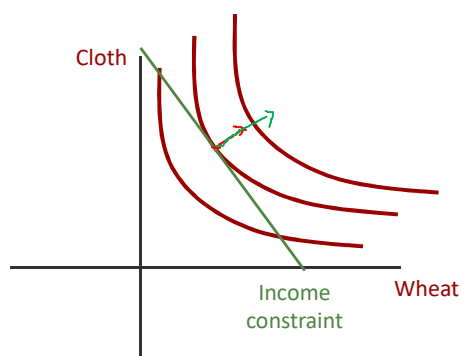
Constrained optimization

- Maximize utility st income constraint

$$U = X_C^a \cdot X_W^b$$

Subject to (s.t.): $Y = P_C \cdot X_C + P_W \cdot X_W$

$$\mathcal{L} = X_C^a \cdot X_W^b - \lambda(P_C \cdot X_C + P_W \cdot X_W - Y)$$



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Constrained Optimization – Lagrangian Method



- Workhorse of economic optimization problems
- Technique is used to find the local minima/maxima of multivariate functions
- **Key:** add/subtract λ (*Lagrange multiplier*) times the constraint. Then differentiate with respect to X_C , X_W and λ . Finally, solve the resulting system of 3 equations

$$U = X_C^a \cdot X_W^b \quad \leftarrow \text{Objective function}$$

$$\text{Subject to (s.t.): } Y = P_C \cdot X_C + P_W \cdot X_W \quad \leftarrow \text{Constraint function}$$

$$\mathcal{L} = X_C^a \cdot X_W^b - \lambda(P_C \cdot X_C + P_W \cdot X_W - Y) \quad \leftarrow \text{Resulting lagrangian function}$$



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Cobb Douglas Demand functions



$$\mathcal{L} = X_C^a \cdot X_W^b - \lambda(P_C \cdot X_C + P_W \cdot X_W - Y)$$

- First order conditions:

$$-\frac{\delta \mathcal{L}}{\delta X_C} = a \cdot X_C^{a-1} \cdot X_W^b - \lambda(P_C) = 0$$

$$-\frac{\delta \mathcal{L}}{\delta X_W} = b \cdot X_C^a \cdot X_W^{b-1} - \lambda(P_W) = 0$$

$$-\frac{\delta \mathcal{L}}{\delta \lambda} = P_C \cdot X_C + P_W \cdot X_W - Y = 0$$



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Cobb Douglas Demand functions



$$\mathcal{L} = X_C^a \cdot X_W^b - \lambda(P_C \cdot X_C + P_W \cdot X_W - Y)$$

- First order conditions:

$$-\frac{\delta \mathcal{L}}{\delta X_C} \Rightarrow a \cdot X_C^{a-1} \cdot X_W^b = \cancel{\lambda}(P_C)$$

$$-\frac{\delta \mathcal{L}}{\delta X_W} \Rightarrow b \cdot X_C^a \cdot X_W^{b-1} = \cancel{\lambda}(P_W)$$

$$-\frac{\delta \mathcal{L}}{\delta \lambda} = P_C \cdot X_C + P_W \cdot X_W - Y = 0$$



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Cobb Douglas Demand functions



$$\frac{MU_C}{MU_W} = \frac{a \cdot X_C^{a-1} \cdot X_W^b}{b \cdot X_C^a \cdot X_W^{b-1}} = \frac{P_C}{P_W}$$

- Simplify LHS: $\frac{a \cdot X_W^b \cdot X_W^{-b+1}}{b \cdot X_C^a \cdot X_C^{-a+1}} = \frac{a \cdot X_W^{b-b+1}}{b \cdot X_C^{a-a+1}}$

$$\frac{a \cdot X_W}{b \cdot X_C} = \frac{P_C}{P_W}$$

- Next rearrange for W and then C to get them as functions of the other:

$$- X_W = \frac{b \cdot P_C}{a \cdot P_W} \cdot X_C$$

$$- X_C = \frac{a \cdot P_W}{b \cdot P_C} \cdot X_W$$



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Cobb Douglas Demand functions



- Then we plug each into the budget constraint
- Plug $X_W = \frac{b \cdot P_C}{a \cdot P_W} \cdot X_C$ into budget constraint to solve for the demand for X_C :
 - $Y = P_C \cdot X_C + P_W \cdot X_W$
 - $Y = P_C \cdot X_C + P_W \cdot \frac{b \cdot P_C}{a \cdot P_W} \cdot X_C$
 - $Y = P_C \cdot X_C \left[1 + P_W \cdot \frac{b}{a \cdot P_W} \right]$
 - $Y = P_C \cdot X_C \cdot \left[1 + \frac{b}{a} \right] = P_C \cdot X_C \cdot \left[\frac{a}{a} + \frac{b}{a} \right]$
 - $X_C = \frac{Y}{P_C} \left[\frac{a}{a+b} \right]$ or $\frac{X_C \cdot P_C}{Y} = \left[\frac{a}{a+b} \right]$
 - This is why a and b are often referred to as the shares.



Cobb Douglas Demand functions



- Plug $X_C = \frac{a \cdot P_W}{b \cdot P_C} \cdot X_W$ into the budget constraint to obtain the demand curve for X_W

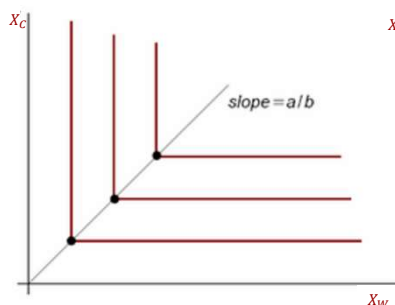
$$\frac{X_W \cdot P_W}{Y} = \left[\frac{b}{a+b} \right]$$

$$\text{or } X_W = \frac{Y}{P_W} \left[\frac{b}{a+b} \right]$$

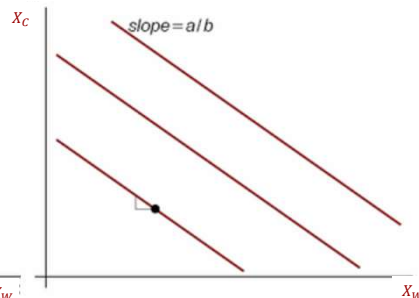
- May also define a composite price to decompose Y into real income and P:
 - $X_C = \frac{X \cdot P}{P_C} \left[\frac{a}{a+b} \right]$ and $X_W = \frac{X \cdot P}{P_W} \left[\frac{b}{a+b} \right]$
 - Emphasizes that it is relative prices that matter. Since $P = S_C \cdot P_C + S_W \cdot P_W$



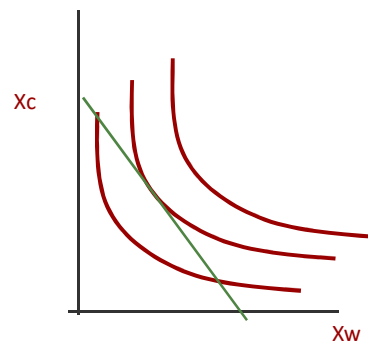
Other demand functions



Leontief fixed proportions
 $U(X_W, X_C) = \min(a \cdot X_W, b \cdot X_C)$



Perfect Substitutes
 $U(X_W, X_C) = a \cdot X_W + b \cdot X_C$



Constant elasticity of substitution
 $U(X_W, X_C) = (X_W^{-\rho} + X_C^{-\rho})^{-1/\rho}$



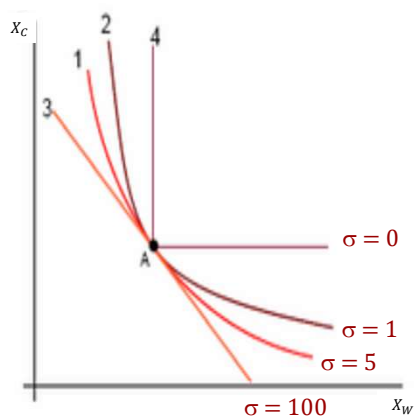
Elasticity

$$\frac{\Delta Y}{\Delta X} \cdot \frac{X}{Y} = \frac{\% \text{ change in } Y}{\% \text{ change in } X}$$

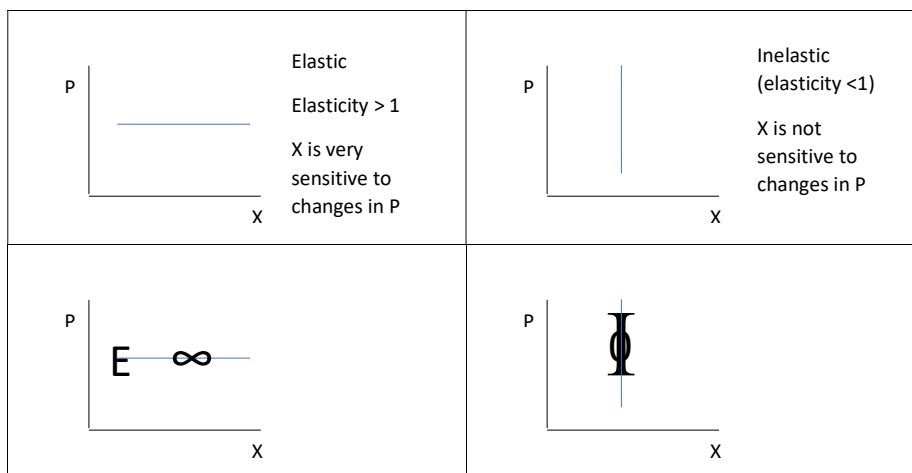
- Sensitivity of Y to changes in X
 - e.g., sensitivity of demand Q to price changes P. When the price of a good increases, we expect demand to fall.
- Linear function: $Q = \beta_0 - \beta_1 \cdot P$
 - elasticity = $\frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q} = \beta_1 \cdot \frac{P}{\beta_0 + \beta_1 \cdot P}$
 - notice that elasticity depends on P and is therefore not constant along a straight line.
- Log functions: $\log(Q) = \beta_0 - \beta_1 \cdot P$
 - elasticity = $\frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q} = \frac{\Delta \log Q}{\Delta \log P} = \beta_1$
 - β_1 is constant
- Cobb Douglas: Elasticity = 1 (prove for yourself)



CES Indifference curves



Elasticity and slopes



Homogeneity

- A function is homogeneous to degree k if:

- $f(s.x_1, s.x_2, s.x_3, \dots, s.x_n) = s^k \cdot f(x_1, x_2, x_3, \dots, x_n)$
- e.g., $Q = A.K^2.L^3$

$$\frac{A.(sK)^2.(sL)^3}{s^2 s^3} = \frac{A.(K)^2.(L)^3}{s^5} \cdot s^5$$

- Is homogenous to degree 5 (increasing returns to scale)
- Other examples:
 - e.g., $Q = A.K^{2/3}.L^{1/3}$ is homogeneous to degree 1 (constant returns to scale)
 - e.g., $Q = A.K^{1/4}.L^{1/4}$ is homogeneous to degree 1/2 (decreasing returns to scale)



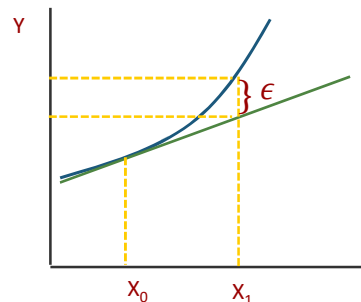
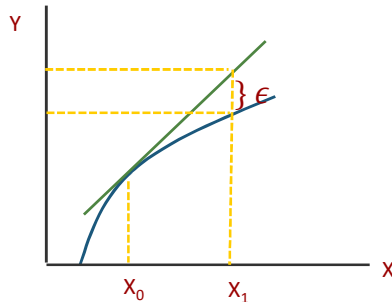
Euler's theorem

- If $f(x)$, where $x \in \mathbb{R}_+^n$ is homogeneous of degree k
 - Then $f_1.x_1 + f_2.x_2 + \dots + f_n.x_n = k.f(x)$
 - Hence if $k = 1$
 - $\sum_{i=1}^n f_i x_i = f(x)$
- Hence for a production function $Q = A.K^{2/3}.L^{1/3}$ that is homogeneous to degree 1:
 - $Q = \frac{dQ}{dK}.K + \frac{dQ}{dL}.L = MPK.K + MPL.L$
 - Which means that all product Q are distributed to the inputs in production (no surplus or deficit)



Taylor Series expansion/approximation

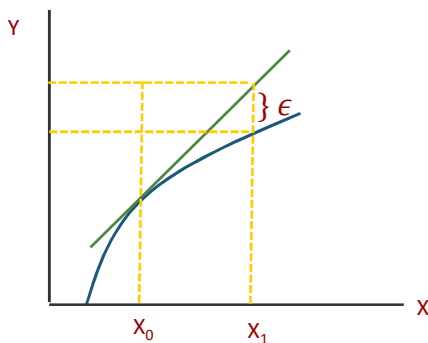
- If we were to use the derivative to find the Δy due to a Δx from x_0 to x_1 . There is likely to be an error ϵ .
 - If function is concave then this method leads to an overestimate
 - If Convex, an underestimate
 - Conversely measuring the error tells us what type of function we have.



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Taylor Series expansion



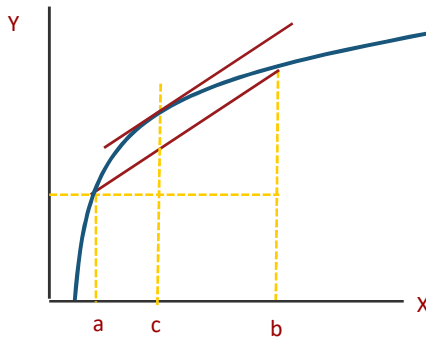
- Taylor series expansion corrects for the error:
 - $$f(x_1) = f(x_0) + \sum_{k=1}^{n-1} \left[\frac{f^{(k)}(x_0) (x_1 - x_0)^k}{k!} \right] + R_n$$
 - Where: R_n is the remainder $R_n = f^{(n)}(\xi)(x_1 - x_0)^n/n!$ and ξ is a value that lies between x_1 and x_0
 - As $k \rightarrow \infty$, the remainder $\rightarrow 0$.
 - Note $k=10$ is usually large enough to get a solution within 6 decimal places.
 - Assuming $k=2$, the function can be rearranged to show that R_n is the error
 - $\Delta y = dy + f^2(\xi)(x_1 - x_0)^2/2$



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Taylor Series expansion



- Assuming $k=1$, we can also obtain the mean value theorem:
 - $f(x_1) = f(x_0) + f'(\xi)(x_1 - x_0)^1$
 - $f'(\xi) = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = f'(x_M)$
- Mean value theorem** is that if f is a continuous function on a closed interval (a, b) then there exists a point c (between a and b) such that the tangent at c is parallel to the secant line through the endpoints, i.e.,
 - $f'(c) = \frac{f(b) - f(a)}{b - a} = f'(c)$



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Envelope Theorem

- Used when our model depends on some endogenous (X_1, X_2) and exogenous (a) variables

$$\mathcal{L} = f(X_1, X_2; a) + \lambda(g(X_1, X_2; a))$$

- Can write a value function: $V(a) = f(X_1(a), X_2(a); a)$

$$\mathcal{L} = f(X_1(a), X_2(a); a) + \lambda(g(X_1(a), X_2(a); a))$$

$$\frac{d\mathcal{L}}{da} = (f_1 + \lambda g_1) \cdot \frac{dX_1}{da} + (f_2 + \lambda g_2) \cdot \frac{dX_2}{da} - g \cdot \frac{d\lambda}{da} + (f_a + \lambda g_a)$$

- At the optimal point $f_i + \lambda g_i = 0$ and $g = 0$, hence:

$$\frac{d\mathcal{L}}{da} = (f_a + \lambda g_a)$$



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